PROJECT REPORT:

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Branch: CSE (AI & ML)

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Course: Artificial Intelligence & Machine Learning

Project Title: Student Performance Prediction & Smart Study Recommendation System

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1. Introduction:

Artificial Intelligence and Machine Learning are transforming various sectors, including education. This project focuses on developing a system that can predict student academic performance and provide personalized study recommendations.

The main goal of this project is to analyze student-related data such as study hours, attendance, sleep patterns, and previous academic scores. Based on this data, the system predicts the student's final performance and provides suggestions to improve their learning outcomes.

This project helps in solving a common issue where students are unaware of their academic standing until exams approach. By using early prediction, students can take necessary actions to improve their performance in time.

2. Objective of the Project:

- To predict student performance using Machine Learning
- To analyze factors affecting academic success
- To provide personalized study recommendations
- To improve students' awareness of their academic progress
- To demonstrate practical application of AI in education

3. Methodology:

In this project, a Machine Learning model is developed using Python. The following steps were followed:

Data Collection

Relevant data such as:

- Study hours
- Attendance percentage
- Sleep duration
- Previous scores

Data Processing

- Data cleaning and formatting using Pandas
- Handling missing values

- Structuring data for model training

Model Development

- Linear Regression algorithm is used
- Model is trained using Scikit-learn
- Predictions are generated based on input data

Recommendation System

A rule-based logic system is implemented:

- Low performance → Suggest more study hours & revision
- Medium performance → Suggest consistency & practice
- High performance → Suggest advanced learning

4. Tools & Technologies Used:

- Programming Language: Python
- Libraries:
- Pandas (data handling)
- Scikit-learn (machine learning model)
- Concepts Used:
- Linear Regression
- Data Analysis
- Predictive Modeling

5. Working of the System:

The system takes input from the user such as study hours, attendance, sleep patterns, and previous scores. This data is passed through the trained Machine Learning model, which predicts the final performance.

Based on the predicted result, the system generates personalized recommendations. These suggestions help students improve their study habits and focus on areas that need attention.

6. Advantages of the Project:

- Helps students understand their performance early
- Provides personalized learning suggestions
- Improves study planning and time management
- Demonstrates real-world use of AI in education
- Easy to use and scalable

7. Challenges Faced

- Selecting relevant features for prediction
- Understanding Machine Learning concepts
- Handling data properly
- Designing meaningful recommendations

Despite these challenges, the project was successfully completed and helped in improving technical knowledge.

8. Conclusion:

This project successfully demonstrates how Artificial Intelligence and Machine Learning can be used in the education sector. By predicting student performance and providing smart recommendations, the system helps students improve their academic outcomes.

It not only enhances learning efficiency but also promotes better decision-making. Overall, this project is a practical example of how technology can support smarter learning strategies.

9. Future Scope

- Adding more features like stress level and extracurricular activities
- Using advanced ML models for better accuracy
- Creating a web or mobile application
- Real-time performance tracking

10. References

- Python Documentation
- Scikit-learn Library
- Pandas Library
- Machine Learning Study Materials